# DocuMind AI Architecture: Cloud-Native Microservices Guide

## Section 1: Distributed RAG System Topology

Modern document intelligence architectures decouple PDF extraction, embedding generation, vector indexing, and generation into discrete, horizontally scalable services. In DocuMind AI, the ingestion engine uses PyMuPDF for structural document extraction.

## Section 2: Embedding and Vector Retrieval

Documents are partitioned into semantic chunks with a configurable token window (typically 512 tokens with 10% overlap). Retrieval uses dense vector search augmented with BM25 hybrid indexing to ensure maximum precision across technical vocabularies.

## Section 3: Telemetry and Verification SLA

Every generated response requires page-level attribution with a target precision threshold of 95%. Responses with grounding scores below 0.70 trigger automated fallback to citation refusal.